

Ankit Yadav  
CS2501

**Q1**

We are given an independent and identically distributed (i.i.d.) sample of  $n$  points  $\{y_i\}$  where each  $y_i \sim N(\mu, 1)$ ,

**(a) Estimator:**  $\hat{\mu} = 1$

$$\text{Bias}(\hat{\mu}) = \mathbb{E}[\hat{\mu}] - \mu$$

$$\text{Since } \hat{\mu} = 1 \text{ is a constant: } \mathbb{E}[\hat{\mu}] = 1 \implies \text{Bias}(\hat{\mu}) = 1 - \mu$$

$$\text{Variance: } \text{Var}(\hat{\mu}) = 0 \quad (\text{variance of a constant is zero})$$

This estimator ignores the data. It is generally not good, because the bias can be arbitrarily large if  $\mu \neq 1$ . However, it may be reasonable if we know  $\mu \approx 1$  from prior information.

**(b) Estimator:**  $\hat{\mu} = y_1$

$$\text{Bias}(\hat{\mu}) = \mathbb{E}[y_1] - \mu \quad \text{Since } y_1 \sim N(\mu, 1): \mathbb{E}[y_1] = \mu \implies \text{Bias}(\hat{\mu}) = 0$$

$$\text{Variance: } \text{Var}(\hat{\mu}) = \text{Var}(y_1) = 1$$

This estimator is unbiased but has high variance, because it uses only one data point. For  $n > 1$ , this is usually not a good estimator. A better choice would be the sample mean, which reduces variance.

**(c)**

$$\text{Consider the estimator } \hat{\mu} = \arg \min_{\mu} \sum_{i=1}^n (y_i - \mu)^2 + \lambda \mu^2, \quad \lambda > 0$$

Take derivative w.r.t  $\mu$  and set to zero:

$$\frac{d}{d\mu} [\sum_{i=1}^n (y_i - \mu)^2 + \lambda \mu^2] = 0$$

$$\sum_{i=1}^n 2(\mu - y_i) + 2\lambda \mu = 0$$

$$n\mu - \sum_{i=1}^n y_i + \lambda\mu = 0$$

$$\mu(n + \lambda) = \sum_{i=1}^n y_i$$

$$\hat{\mu} = \frac{\sum_{i=1}^n y_i}{n+\lambda} = \frac{n}{n+\lambda} \bar{y} \text{ where } \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \text{ is the usual sample mean.}$$

$$\text{Bias}(\hat{\mu}) = \mathbb{E}[\hat{\mu}] - \mu = \frac{n}{n+\lambda} \mathbb{E}[\bar{y}] - \mu$$

$$\mathbb{E}[\bar{y}] = \mu \implies \text{Bias}(\hat{\mu}) = \frac{n}{n+\lambda} \mu - \mu = -\frac{\lambda}{n+\lambda} \mu$$

$$\text{Var}(\hat{\mu}) = \text{Var}\left(\frac{n}{n+\lambda} \bar{y}\right) = \left(\frac{n}{n+\lambda}\right)^2 \text{Var}(\bar{y})$$

$$\text{Var}(\bar{y}) = \frac{1}{n} \implies \text{Var}(\hat{\mu}) = \frac{n}{(n+\lambda)^2}$$

## Q2

We have to prove:  $\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{kj})^2$ ,

$$\text{where } \bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}$$

### LHS

$$\sum_{i,i'} (x_{ij} - x_{i'j})^2 = \sum_{i,i'} (x_{ij}^2 + x_{i'j}^2 - 2x_{ij}x_{i'j})$$

$$= \sum_{i,i'} x_{ij}^2 + \sum_{i,i'} x_{i'j}^2 - 2 \sum_{i,i'} x_{ij}x_{i'j}$$

$$\text{As } \sum_{i,i'} x_{ij}^2 = |C_k| \sum_i x_{ij}^2, \quad \sum_{i,i'} x_{i'j}^2 = |C_k| \sum_{i'} x_{i'j}^2$$

$$\text{So, } \sum_{i,i'} (x_{ij} - x_{i'j})^2 = 2|C_k| \sum_i x_{ij}^2 - 2 \sum_{i,i'} x_{ij}x_{i'j}$$

$$\text{Now, } \sum_{i,i'} x_{ij}x_{i'j} = (\sum_i x_{ij})^2$$

$$\text{So, } \sum_{i,i'} (x_{ij} - x_{i'j})^2 = 2|C_k| \sum_i x_{ij}^2 - 2 (\sum_i x_{ij})^2$$

$$\text{Divide by } |C_k|, \quad \frac{1}{|C_k|} \sum_{i,i'} (x_{ij} - x_{i'j})^2 = 2 \sum_i x_{ij}^2 - \frac{2}{|C_k|} (\sum_i x_{ij})^2$$

### RHS

$$\sum_i (x_{ij} - \bar{x}_{kj})^2 = \sum_i (x_{ij}^2 - 2x_{ij}\bar{x}_{kj} + \bar{x}_{kj}^2)$$

$$= \sum_i x_{ij}^2 - 2\bar{x}_{kj} \sum_i x_{ij} + |C_k| \bar{x}_{kj}^2$$

$$\text{Since, } \bar{x}_{kj} = \frac{1}{|C_k|} \sum_i x_{ij}$$

we get,  $\sum_i (x_{ij} - \bar{x}_{kj})^2 = \sum_i x_{ij}^2 - \frac{1}{|C_k|} (\sum_i x_{ij})^2$

Multiply both sides by 2:  $2 \sum_i (x_{ij} - \bar{x}_{kj})^2 = 2 \sum_i x_{ij}^2 - \frac{2}{|C_k|} (\sum_i x_{ij})^2$

Thus, for each feature  $j$ :  $\frac{1}{|C_k|} \sum_{i,i'} (x_{ij} - x_{i'j})^2 = 2 \sum_i (x_{ij} - \bar{x}_{kj})^2$

Summing over  $j = 1, \dots, p$ :

$$\boxed{\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{kj})^2}$$

### Q3

We are given the objective function  $\min \sum_{k=1}^K \sum_{i \in C_k} \sum_{t=1}^p w_t^\beta (x_{it} - v_{kt})^2$ , subject to  $\sum_{t=1}^p w_t = 1$ ,  $0 \leq w_t \leq 1$ ,  $\beta > 1$

#### Update for Cluster Centers $v_k$

Fix  $C_k$  and  $w_t$ , minimize w.r.t.  $v_{kt}$ :

$J = \sum_{i \in C_k} \sum_{t=1}^p w_t^\beta (x_{it} - v_{kt})^2$  Take derivative w.r.t.  $v_{kt}$ :

$$\frac{\partial J}{\partial v_{kt}} = \sum_{i \in C_k} 2w_t^\beta (v_{kt} - x_{it}) = 0$$

$$\Rightarrow \sum_{i \in C_k} (v_{kt} - x_{it}) = 0$$

$$|C_k| v_{kt} = \sum_{i \in C_k} x_{it}$$

$$v_{kt} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{it}$$

#### Update for Cluster Assignments $C_k$

Fix  $v_k$  and  $w_t$ . Each point  $x_i$  is assigned to the cluster minimizing:

$$\sum_{t=1}^p w_t^\beta (x_{it} - v_{kt})^2$$

$$\text{Thus, } C_k = \left\{ i : k = \arg \min_j \sum_{t=1}^p w_t^\beta (x_{it} - v_{jt})^2 \right\}$$

#### Update for Feature Weights $w_t$

Fix  $C_k$  and  $v_k$ . Define:  $D_t = \sum_{k=1}^K \sum_{i \in C_k} (x_{it} - v_{kt})^2$

The objective becomes,  $\min_{w_t} \sum_{t=1}^p w_t^\beta D_t$  s.t.  $\sum_{t=1}^p w_t = 1$

Using Lagrange multipliers:

$$\mathcal{L} = \sum_{t=1}^p w_t^\beta D_t + \lambda (\sum_{t=1}^p w_t - 1)$$

Take derivative w.r.t.  $w_t$ :  $\frac{\partial \mathcal{L}}{\partial w_t} = \beta w_t^{\beta-1} D_t + \lambda = 0$

$$\Rightarrow w_t^{\beta-1} = -\frac{\lambda}{\beta D_t} \text{ Let } c = -\frac{\lambda}{\beta}:$$

$$w_t^{\beta-1} = \frac{c}{D_t}$$

$$w_t = \left( \frac{c}{D_t} \right)^{\frac{1}{\beta-1}}$$

Use constraint  $\sum_t w_t = 1$ :

$$\sum_{t=1}^p \left( \frac{c}{D_t} \right)^{\frac{1}{\beta-1}} = 1$$

$$c^{\frac{1}{\beta-1}} \sum_{t=1}^p D_t^{-\frac{1}{\beta-1}} = 1$$

$$c^{\frac{1}{\beta-1}} = \frac{1}{\sum_{s=1}^p D_s^{-\frac{1}{\beta-1}}}$$

Substituting it back, we get:

$$w_t = \frac{D_t^{-\frac{1}{\beta-1}}}{\sum_{s=1}^p D_s^{-\frac{1}{\beta-1}}}$$

## Q4

We are given the log-likelihood,  $LL(\theta) = \sum_{i=1}^m [y^{(i)} \log(h_\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)}))]$

, where  $h_\theta(x^{(i)}) = g(\theta^T x^{(i)})$ ,  $g(z) = \frac{1}{1+e^{-z}}$

We have to show that  $LL(\theta)$  is **concave** in  $\theta$ .

### Rewrite the Log-Likelihood

Using the identity:  $g(z) = \frac{1}{1+e^{-z}}$ ,  $1 - g(z) = \frac{e^{-z}}{1+e^{-z}}$

$$\log(g(z)) = -\log(1 + e^{-z})$$

$$\log(1 - g(z)) = -z - \log(1 + e^{-z})$$

Substitute into  $LL(\theta)$  with  $z_i = \theta^T x^{(i)}$ :

$$LL(\theta) = \sum_{i=1}^m [y^{(i)}(-\log(1 + e^{-z_i})) + (1 - y^{(i)})(-z_i - \log(1 + e^{-z_i}))]$$

Simplify, we get:  $LL(\theta) = \sum_{i=1}^m [-\log(1 + e^{-z_i}) - (1 - y^{(i)})z_i]$

Rearranging:

$$LL(\theta) = \sum_{i=1}^m [y^{(i)}z_i - \log(1 + e^{z_i})]$$

Thus,  $LL(\theta) = \sum_{i=1}^m [y^{(i)}\theta^T x^{(i)} - \log(1 + e^{\theta^T x^{(i)}})]$

### Compute the Gradient

$$\begin{aligned}\nabla LL(\theta) &= \sum_{i=1}^m \left[ y^{(i)}x^{(i)} - \frac{e^{\theta^T x^{(i)}}}{1 + e^{\theta^T x^{(i)}}}x^{(i)} \right] \\ &= \sum_{i=1}^m [(y^{(i)} - g(\theta^T x^{(i)}))x^{(i)}]\end{aligned}$$

### Compute the Hessian

Differentiate again:  $\nabla^2 LL(\theta) = -\sum_{i=1}^m g(\theta^T x^{(i)})(1 - g(\theta^T x^{(i)}))x^{(i)}x^{(i)T}$

Let,  $\sigma_i = g(\theta^T x^{(i)})$ ,  $\sigma_i(1 - \sigma_i) > 0$

Thus,  $\nabla^2 LL(\theta) = -\sum_{i=1}^m \sigma_i(1 - \sigma_i)x^{(i)}x^{(i)T}$

### Show Negative Semi-Definiteness

For any vector  $z \in \mathbb{R}^n$ :  $z^T \nabla^2 LL(\theta) z = -\sum_{i=1}^m \sigma_i(1 - \sigma_i)(z^T x^{(i)})^2$

Since,  $\sigma_i(1 - \sigma_i) > 0$ ,  $(z^T x^{(i)})^2 \geq 0$

we get,  $z^T \nabla^2 LL(\theta) z \leq 0$

Thus, the Hessian is **negative semi-definite**.

$\nabla^2 LL(\theta) \leq 0 \Rightarrow LL(\theta)$  is concave in  $\theta$ .

## Q5

Let  $x_1, x_2, \dots, x_n \in \mathbb{R}^d$  and let  $A \in \mathbb{R}^{d \times d}$  be a non-singular matrix. Consider the objective:  
 $J(x) = \sum_{k=1}^n (x_k - x)^T A^{-1} (x_k - x)$

We show that the minimizer is:  $x^* = \frac{1}{n} \sum_{k=1}^n x_k$

### Expand the Objective function

Expand each term:  $(x_k - x)^T A^{-1}(x_k - x) = x_k^T A^{-1} x_k - 2x^T A^{-1} x_k + x^T A^{-1} x$

Summing over  $k$ :  $J(x) = \sum_{k=1}^n x_k^T A^{-1} x_k - 2 \sum_{k=1}^n x^T A^{-1} x_k + \sum_{k=1}^n x^T A^{-1} x$

Simplify:  $J(x) = \sum_{k=1}^n x_k^T A^{-1} x_k - 2x^T A^{-1} \sum_{k=1}^n x_k + nx^T A^{-1} x$

### Compute the Gradient

Differentiate  $J(x)$  w.r.t.  $x$ :  $\nabla_x J(x) = -2A^{-1} \sum_{k=1}^n x_k + 2nA^{-1}x$

### Set Gradient to Zero

$$-2A^{-1} \sum_{k=1}^n x_k + 2nA^{-1}x = 0$$

Divide by 2, we get:  $-A^{-1} \sum_{k=1}^n x_k + nA^{-1}x = 0$

Multiply both sides by  $A$ :  $-\sum_{k=1}^n x_k + nx = 0$

$$nx = \sum_{k=1}^n x_k$$

$$x = \frac{1}{n} \sum_{k=1}^n x_k$$

### Verify Minimum

The Hessian is:  $\nabla^2 J(x) = 2nA^{-1}$

Since  $A$  is non-singular,  $A^{-1}$  exists.

If  $A$  is positive definite (typical in such problems), then  $A^{-1}$  is also positive definite, implying:  $\nabla^2 J(x) > 0$

So  $J(x)$  is strictly convex and the solution is a **global minimum**.

$$\text{So, } \arg \min_x \sum_{k=1}^n (x_k - x)^T A^{-1} (x_k - x) = \frac{1}{n} \sum_{k=1}^n x_k$$

### Q7

We are given the linear model  $Y = X\beta + \epsilon$  where  $\epsilon_i \sim \text{Laplace}(0, b)$  i.i.d., with density  $p(\epsilon_i) = \frac{1}{2b} \exp\left(-\frac{|\epsilon_i|}{b}\right)$ .

Since  $\epsilon_i = y_i - x_i^T \beta$ , we can write  $p(y_i | x_i, \beta) = \frac{1}{2b} \exp\left(-\frac{|y_i - x_i^T \beta|}{b}\right)$

### (a) Derivation of the Loss Function

The likelihood over all data points is  $P(Y | X, \beta) = \prod_{i=1}^m \frac{1}{2b} \exp\left(-\frac{|y_i - x_i^T \beta|}{b}\right)$

$$\text{Log-likelihood: } \log P(Y | X, \beta) = \sum_{i=1}^m \left[ \log \frac{1}{2b} - \frac{|y_i - x_i^T \beta|}{b} \right]$$

$$= -m \log(2b) - \frac{1}{b} \sum_{i=1}^m |y_i - x_i^T \beta|$$

As  $-m \log(2b)$  is constant w.r.t.  $\beta$ , maximizing log-likelihood is equivalent to minimizing:

$$J_{\text{Laplace}}(\beta) = \sum_{i=1}^m |y_i - x_i^T \beta|$$

Thus, the MLE under Laplace noise corresponds to **minimizing the  $\ell_1$  loss**.

### (b) Why this is more robust?

#### 1. Comparison of Loss Functions

- Gaussian noise  $\Rightarrow$  least squares loss:  $\sum_{i=1}^m (y_i - x_i^T \beta)^2$
- Laplace noise  $\Rightarrow$  absolute loss:  $\sum_{i=1}^m |y_i - x_i^T \beta|$

#### 2. Sensitivity to Outliers

- **Squared loss** grows quadratically: If residual  $r \rightarrow \infty$ ,  $r^2 \rightarrow \infty$  very fast  
Thus, large errors dominate the objective.
- **Absolute loss** grows linearly:  $|r| \rightarrow \infty$  linearly  
Thus, large residuals have less influence.

#### 3. Gradient Behavior

- Squared loss gradient:  $\frac{d}{dr}(r^2) = 2r$  (increases with  $r$ )
- Absolute loss subgradient:  $\frac{d}{dr}|r| = \text{sign}(r)$  (bounded)

This means:

- Large errors produce very large gradients in least squares  $\Rightarrow$  strong influence
- Large errors produce bounded gradients in  $\ell_1$  loss  $\Rightarrow$  limited influence

#### 4. Heavy-Tailed Noise Interpretation

- Gaussian distribution has **light tails**
- Laplace distribution has **heavier tails**

Thus:

- Laplace model assigns higher probability to large deviations
- It better accommodates outliers in data

## Q8

We are given the dataset:

| Gender   | Income      | Approved | Count |
|----------|-------------|----------|-------|
| <i>F</i> | <i>Low</i>  | +        | 10    |
| <i>F</i> | <i>High</i> | +        | 95    |
| <i>M</i> | <i>Low</i>  | +        | 5     |
| <i>M</i> | <i>High</i> | +        | 90    |
| <i>F</i> | <i>Low</i>  | −        | 80    |
| <i>F</i> | <i>High</i> | −        | 20    |
| <i>M</i> | <i>Low</i>  | −        | 120   |
| <i>M</i> | <i>High</i> | −        | 30    |

### (a) Entropy $H(\text{Approved})$

Total positive:  $10 + 95 + 5 + 90 = 200$

Total negative:  $80 + 20 + 120 + 30 = 250$

Total samples: 450

$$p_+ = \frac{200}{450} = \frac{4}{9}, \quad p_- = \frac{250}{450} = \frac{5}{9}$$

Entropy:  $H(\text{Approved}) = -p_+ \log_2 p_+ - p_- \log_2 p_-$

$$= -\frac{4}{9} \log_2 \frac{4}{9} - \frac{5}{9} \log_2 \frac{5}{9}$$

$$H(\text{Approved}) \approx -\frac{4}{9}(-1.17) - \frac{5}{9}(-0.85)$$

$$\approx 0.52 + 0.47 = 0.99$$

$H(\text{Approved}) \approx 0.99$

### (b) Information Gain

#### 1. $IG(\text{Gender})$

**For Gender = F:**

Positive:  $10 + 95 = 105$

Negative:  $80 + 20 = 100$

$$p_+ = \frac{105}{205}, \quad p_- = \frac{100}{205}$$

$$H_F \approx -0.512 \log_2(0.512) - 0.488 \log_2(0.488) \approx 1.00$$

**For Gender = M:**

Positive:  $5 + 90 = 95$



Negative:  $120 + 30 = 150$

$$p_+ = \frac{95}{245}, \quad p_- = \frac{150}{245}$$

$$H_M \approx -0.388 \log_2(0.388) - 0.612 \log_2(0.612) \approx 0.96$$

Weighted entropy:

$$H(\text{Approved}|\text{Gender}) = \frac{205}{450}(1.00) + \frac{245}{450}(0.96)$$

$$\approx 0.456 + 0.523 = 0.979$$

$$IG(\text{Gender}) = 0.99 - 0.979 = 0.011$$

$IG(\text{Gender}) \approx 0.011$

## 2. $IG(\text{Income})$

**For Income = Low:**

Positive:  $10 + 5 = 15$

Negative:  $80 + 120 = 200$

$$p_+ = \frac{15}{215}, \quad p_- = \frac{200}{215}$$

$$H_{Low} \approx -0.07 \log_2(0.07) - 0.93 \log_2(0.93) \approx 0.37$$

**For Income = High:**

Positive:  $95 + 90 = 185$

Negative:  $20 + 30 = 50$

$$p_+ = \frac{185}{235}, \quad p_- = \frac{50}{235}$$

$$H_{High} \approx -0.787 \log_2(0.787) - 0.213 \log_2(0.213) \approx 0.75$$

Weighted entropy:

$$H(\text{Approved}|\text{Income}) = \frac{215}{450}(0.37) + \frac{235}{450}(0.75)$$

$$\approx 0.177 + 0.392 = 0.569$$

$$IG(\text{Income}) = 0.99 - 0.569 = 0.421$$

$IG(\text{Income}) \approx 0.421$

## (c) Decision Tree (ID3)

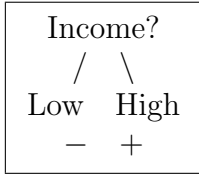
Since:  $IG(\text{Income}) > IG(\text{Gender})$ , the root node is **Income**.

### Split on Income

- Income = Low:
  - Positive = 15, Negative = 200  $\Rightarrow$  majority class = -
- Income = High:
  - Positive = 185, Negative = 50  $\Rightarrow$  majority class = +

Thus, no further split is needed.

### Final Decision Tree



## Q10

We are given a mixture of two 1D Gaussians with unit variance:

$$P(x; \mu, \theta) = \sum_{y \in \{0,1\}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu_y)^2}{2}\right) \theta_y,$$

where:  $\theta_y = P(Y = y), \quad \theta_0 + \theta_1 = 1$

Let the dataset be  $\{x_i\}_{i=1}^n$ .

(a) **E-step: Compute**  $p_{iy} = P(Y = y \mid X = x_i)$

By Bayes' rule:  $p_{iy} = \frac{P(x_i|y) P(y)}{\sum_{y' \in \{0,1\}} P(x_i|y') P(y')}$

Substitute the Gaussian density:  $P(x_i \mid y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu_y)^2}{2}\right)$

Canceling the common constant  $\frac{1}{\sqrt{2\pi}}$ , we get:

$$p_{iy} = \frac{\theta_y \exp\left(-\frac{(x_i - \mu_y)^2}{2}\right)}{\sum_{y' \in \{0,1\}} \theta_{y'} \exp\left(-\frac{(x_i - \mu_{y'})^2}{2}\right)}$$

Thus, at iteration  $t$ :

|                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $p_{iy}^{(t)} = \frac{\theta_y^{(t-1)} \exp\left(-\frac{(x_i - \mu_y^{(t-1)})^2}{2}\right)}{\theta_0^{(t-1)} \exp\left(-\frac{(x_i - \mu_0^{(t-1)})^2}{2}\right) + \theta_1^{(t-1)} \exp\left(-\frac{(x_i - \mu_1^{(t-1)})^2}{2}\right)}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**(b) M-step: Update  $\mu$  and  $\theta$**

We treat  $p_{iy}$  as fractional counts.

**Update for  $\mu_y$**

We maximize:  $\sum_{i=1}^n \sum_y p_{iy} \log P(x_i | y)$

Ignoring constants:  $\sum_{i=1}^n p_{iy} \left( -\frac{(x_i - \mu_y)^2}{2} \right)$

This is equivalent to minimizing:  $\sum_{i=1}^n p_{iy} (x_i - \mu_y)^2$

Taking derivative:

$$\begin{aligned} \frac{\partial}{\partial \mu_y} \sum_i p_{iy} (x_i - \mu_y)^2 &= 0 \\ \sum_i p_{iy} (\mu_y - x_i) &= 0 \end{aligned}$$

$$\mu_y \sum_i p_{iy} = \sum_i p_{iy} x_i$$

$$\boxed{\mu_y^{(t)} = \frac{\sum_{i=1}^n p_{iy}^{(t)} x_i}{\sum_{i=1}^n p_{iy}^{(t)}}}$$

**Update for  $\theta_y$**

We maximize:  $\sum_{i=1}^n \sum_y p_{iy} \log \theta_y$

subject to:  $\theta_0 + \theta_1 = 1$

Using Lagrange multipliers:  $\theta_y \propto \sum_{i=1}^n p_{iy}$

$$\text{Normalize: } \boxed{\theta_y^{(t)} = \frac{1}{n} \sum_{i=1}^n p_{iy}^{(t)}}$$

**Q6**

We are given a Gaussian Mixture Model:  $gmm(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k)$

with  $\sum_{k=1}^K \pi_k = 1$ ,  $\pi_k \geq 0$ .

Let  $z_{ik} := \tilde{z}_{ik} = P(\text{cluster } k | x_i)$ .

**(a) Update for  $\Sigma$  when  $\Sigma_k = \Sigma$**

We have to maximize:  $\sum_{i=1}^n \sum_{k=1}^K z_{ik} \log \mathcal{N}(x_i | \mu_k, \Sigma)$

Gaussian log-density:  $\log \mathcal{N}(x_i | \mu_k, \Sigma) = -\frac{1}{2} \log |\Sigma| - \frac{1}{2} (x_i - \mu_k)^T \Sigma^{-1} (x_i - \mu_k) + C$

Thus objective:  $L = -\frac{1}{2} \sum_{i,k} z_{ik} [\log |\Sigma| + (x_i - \mu_k)^T \Sigma^{-1} (x_i - \mu_k)]$

**Derivative w.r.t.  $\Sigma^{-1}$ :**

$$\frac{\partial}{\partial \Sigma^{-1}} \log |\Sigma| = -\Sigma$$

$$\frac{\partial}{\partial \Sigma^{-1}} (x_i - \mu_k)^T \Sigma^{-1} (x_i - \mu_k) = (x_i - \mu_k)(x_i - \mu_k)^T$$

Set derivative to zero:

$$\sum_{i,k} z_{ik} [\Sigma - (x_i - \mu_k)(x_i - \mu_k)^T] = 0$$

$$\Sigma \sum_{i,k} z_{ik} = \sum_{i,k} z_{ik} (x_i - \mu_k)(x_i - \mu_k)^T$$

$$\text{As } \sum_k z_{ik} = 1: \sum_{i,k} z_{ik} = n$$

$$\text{So, } \boxed{\Sigma = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K z_{ik} (x_i - \mu_k)(x_i - \mu_k)^T}$$

**(b) As  $\sigma^2 \rightarrow 0$ , EM  $\rightarrow$  Lloyd's algorithm**

We have,  $P(x_i | k) = \mathcal{N}(x_i | \mu_k, \sigma^2 I)$

$$\text{E-step: } z_{ik} = \frac{\pi_k \exp\left(-\frac{1}{2\sigma^2} \|x_i - \mu_k\|^2\right)}{\sum_j \pi_j \exp\left(-\frac{1}{2\sigma^2} \|x_i - \mu_j\|^2\right)}$$

**As  $\sigma^2 \rightarrow 0$ :**

- The exponent  $\exp(-\|x_i - \mu_k\|^2 / (2\sigma^2))$  becomes:

- Dominated by the smallest distance.
- Goes to 0 for all other clusters

$$\text{So, } z_{ik} \rightarrow \begin{cases} 1 & \text{if } k = \arg \min_j \|x_i - \mu_j\|^2 \\ 0 & \text{otherwise} \end{cases}$$

As  $\sigma^2 \rightarrow 0$ , EM assignments become hard and reduce to Lloyd's algorithm.

**(c) EM as Gradient Ascent**

We are given gradient ascent updates:

$$\mu_k^{(t+1)} = \mu_k^{(t)} + \eta_k^{(t)} \nabla_{\mu_k} l$$

$$(\sigma_k^2)^{(t+1)} = (\sigma_k^2)^{(t)} + s_k^{(t)} \nabla_{\sigma_k^2} l$$

**Gradient w.r.t.  $\mu_k$**

$$\text{As, } \nabla_{\mu_k} l = \sum_{i=1}^n z_{ik} \frac{(x_i - \mu_k)}{\sigma_k^2}$$

$$\text{Choose step size: } \eta_k^{(t)} = \frac{\sigma_k^2}{\sum_i z_{ik}}$$

$$\text{Then: } \mu_k^{(t+1)} = \mu_k^{(t)} + \frac{\sigma_k^2}{\sum_i z_{ik}} \sum_i z_{ik} \frac{(x_i - \mu_k)}{\sigma_k^2}$$

$$= \mu_k^{(t)} + \frac{1}{\sum_i z_{ik}} \sum_i z_{ik} (x_i - \mu_k)$$

$$= \frac{\sum_i z_{ik} x_i}{\sum_i z_{ik}}$$

$$\boxed{\mu_k^{(t+1)} = \frac{\sum_i z_{ik} x_i}{\sum_i z_{ik}}}$$

This matches the EM M-step for  $\mu_k$ .

**Gradient w.r.t.  $\sigma_k^2$**

$$\nabla_{\sigma_k^2} l = \sum_i z_{ik} \left[ -\frac{d}{2\sigma_k^2} + \frac{\|x_i - \mu_k\|^2}{2\sigma_k^4} \right]$$

$$\text{Choose step size: } s_k^{(t)} = \frac{2(\sigma_k^2)^2}{\sum_i z_{ik}}$$

$$\text{So, } \boxed{\sigma_k^2 = \frac{\sum_i z_{ik} \|x_i - \mu_k\|^2}{d \sum_i z_{ik}}}$$

With appropriate step sizes, gradient ascent updates matches with EM updates.

## Q9

We are given  $J(\gamma, \{\mu_k\}) = \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik} \|x_i - \mu_k\|^2$   
 where  $\gamma_{ik} \in \{0, 1\}$  and  $\sum_{k=1}^K \gamma_{ik} = 1$ .

### (a) Finite termination of Lloyd's algorithm

Each  $\gamma$  represents an assignment of  $n$  points to  $K$  clusters.

Number of possible assignments:  $K^n$

At each iteration:

- Assignment step reduces  $J$  (or keeps it unchanged)
- Update step minimizes  $J$  for fixed assignments

Thus:

- $J$  is monotonically non-increasing
- Only finitely many  $\gamma$  exist

Therefore, the algorithm cannot cycle indefinitely.

Lloyd's algorithm terminates in a finite number of steps

**(b) Relation between total, within, and between variations**

Define:  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

**Total variation:**  $V(X) = \frac{1}{n} \sum_{i=1}^n \|x_i - \bar{x}\|^2$

**Within-cluster variation:**  $V_k(X) = \frac{\sum_i \gamma_{ik} \|x_i - \mu_k\|^2}{\sum_i \gamma_{ik}}$

**Between-cluster variation:**  $\tilde{V}(X) = \sum_{k=1}^K \left( \frac{\sum_i \gamma_{ik}}{n} \right) \|\mu_k - \bar{x}\|^2$

Using,  $\|x_i - \bar{x}\|^2 = \|x_i - \mu_k\|^2 + \|\mu_k - \bar{x}\|^2 + 2(x_i - \mu_k)^T(\mu_k - \bar{x})$

Summing over  $i$  in cluster  $k$ , the cross term vanishes:  $\sum_i \gamma_{ik}(x_i - \mu_k) = 0$

So,  $\sum_i \|x_i - \bar{x}\|^2 = \sum_k \sum_i \gamma_{ik} \|x_i - \mu_k\|^2 + \sum_k \sum_i \gamma_{ik} \|\mu_k - \bar{x}\|^2$

Divide by  $n$ :  $V(X) = \sum_k \frac{n_k}{n} V_k(X) + \tilde{V}(X)$ , where  $n_k = \sum_i \gamma_{ik}$ .

**Interpretation:**

- Total variance = within-cluster variance + between-cluster variance
- Since  $V(X)$  is fixed, minimizing within-cluster variance maximizes between-cluster variance

K-means minimizes within-cluster variation while maximizing separation

**(c) Manhattan distance version**

Objective:  $J_1 = \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik} \|x_i - \mu_k\|_1$

We have to minimize:  $\sum_i \gamma_{ik} \|x_i - \mu_k\|_1$

Since:  $\|x_i - \mu_k\|_1 = \sum_{j=1}^d |x_{ij} - \mu_{kj}|$

So,  $\min_{\mu_{kj}} \sum_i \gamma_{ik} |x_{ij} - \mu_{kj}|$

Using the given fact:  $\min_x \sum_i |x - r_i| \Rightarrow x = \text{median}(r_i)$

So,  $\mu_{kj} = \text{median}\{x_{ij} : \gamma_{ik} = 1\}$

So each coordinate of  $\mu_k$  is the median of that coordinate in cluster  $k$ .

**(d) Monotonicity in  $K$**

Let  $J_K$  be the minimum distortion with  $K$  clusters.

Consider  $K + 1$  clusters:

- Any  $K$ -cluster solution is also valid for  $K + 1$  clusters (just leave one cluster empty)

So,  $J_{K+1} \leq J_K$

$J_K$  is non-increasing in  $K$

So,

- Increasing  $K$  always reduces (or keeps same) distortion
- Minimum occurs at  $K = n$  (each point its own cluster)

So, minimizing  $J$  alone cannot determine optimal  $K$